UBT406: BIOANALYTICAL TECHNIQUES

L	T	P	Cr
3	0	0	3.0

Course Objective: The objectives of this course are to provide the students with the understanding of various analytical techniques used in biotechnology-based research and industry. The course will acquaint the students with the various instruments, their configuration and principle of working, operating procedures, data generation and its analysis.

Syllabus

Sampling and sample preparation: Sample fixing for various analytical applications and sample processing

Introduction to chromatographic techniques: Theoretical basis of chromatographic separations. Column, thin layer, Paper, Normal phase and reverse phase chromatography, Ion-exchange, Affinity and Gas Chromatography, High performance liquid chromatography (HPLC)

Electrophoretic techniques: Theory and application of polyacrylamide and agarose gel electrophoresis, electrophoresis of protein and nucleic acids, Capillary electrophoresis

Centrifugation techniques: Introduction, Basic principle of sedimentation, Centrifuges and their uses, safety aspects in the use of centrifuges. Density gradient and analytical centrifugation

Spectroscopic techniques: Theory and application of UV-VIS, IR, NMR, Fluorescence, Atomic absorption spectroscopy; X-ray diffraction. Introduction to mass spectroscopy

Radioisotopic techniques: Introduction to radioisotopes, detection, measurement and uses of radioisotopes, counting efficiency and autoradiography, biotechnological applications

Microscopy: Principles of microscopy, Light, dark field, fluorescent, UV, transmission and Scanning electron microscopy, Confocal microscopy, microtomy and analysis and measurement of images

Laboratory Work (if applicable)

Course Learning Objectives (CLO)
The students will be able to:

1. Apply basic principles of different analytical techniques in analytical work.

2. Use spectroscopy and radioactivity in biotechnological applications.
3. Use microscopy, centrifugation and electrophoretic techniques.
4. Demonstrate principle and working of various instruments.
5. Use various techniques for solving industrial and research problems.

Text Books

1. Wilson K., Walker J. Principle and Techniques of Biochemistry and Molecular Biology. Cambridge University Press (2010) 7th edition.
2. Pingoud A., Urbanke C., et al. Biochemical Methods – A concise guide for students and researchers. Wiley (2002).
3. Stryer, A.L., Berg J.A. and Tymoczko, J.L., Biochemistry, W.H.Freeman & Co Ltd (2002).

Reference Books

1. Hawes C., Satiat-Jeunemaitre B. Plant Cell Biology. Oxford University Press (2001) 2nd edition.
2. McHale J.L. Molecular Spectroscopy. Pearson (2008) 1st edition.
3. Zubey, G.L., Principles of Biochemistry, Pearson-Education (2007).
4. Marimuthu R. Microscopy and Microtechniques. MJP Publishers Chennai (2008).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/quizzes)	30